

Quantification: 542_z18

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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image

measure	value
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lacuna count	12
interior lacuna count	11
frame-edge lacuna count	1
gap bridges added (detection)	14

Interior lacunae only: mean, median and sample SD (n = 11)

measure	kind	mean	median	sd	units
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area_px2	lacuna shape	3688.27	3684	1441.75	px^2
major_axis_length_px	lacuna shape	132.275	128.431	35.9804	px
minor_axis_length_px	lacuna shape	35.8241	35.4482	8.8535	px
aspect_ratio	lacuna shape	3.6839	3.6773	0.6253	unitless
eccentricity	lacuna shape	0.9593	0.9623	0.0137	unitless
solidity	lacuna shape	0.8643	0.893	0.0736	unitless
roots_count	headline	8.2727	8	4.0023	count
ring_length_r30_px	headline	357.091	376	125.504	px
ring_length_r60_px	reported	968.727	1007	301.731	px
owned_length_px	ownership-dependent	972.615	961.477	477.436	px
edge_count	ownership-dependent	32	33	17.2743	count
mean_edge_length_px	ownership-dependent	30.7951	31.5344	5.1473	px
perimeter_px	normalised	349.688	337.706	124.859	px
ring_area_r30_px2	normalised	12006.2	12118	2958.11	px^2
ring_area_r60_px2	normalised	28554.3	29514	6304.69	px^2
in_frame_fraction_r30	normalised	0.9834	1	0.0361	unitless
in_frame_fraction_r60	normalised	0.9555	1	0.0828	unitless
ring_density_r30	normalised	0.0290011	0.0285614	0.00576019	px^-1
ring_density_r60	normalised	0.0332627	0.0340979	0.0063714	px^-1
roots_per_100px_perimeter	normalised	2.3636	2.2772	0.4767	per 100 px
ring_attached_length_r30_px	network v2	239.909	231	115.754	px
ring_attached_length_r60_px	network v2	479.818	438	237.446	px
ring_length_w_r30_px	network v2	397.173	421.563	144.959	px
ring_length_w_r60_px	network v2	1090.32	1145.64	345.107	px
sholl_crossings_r10	network v2	9.3636	9	4.6534	count
sholl_crossings_r20	network v2	13.2727	13	4.7559	count
sholl_crossings_r30	network v2	14.8182	16	5.115	count
ring_width_mean_r30_px	width	5.5062	5.5614	0.5316	px

Per field

measure	value	units
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analysed_area_px2	1.0074e+06	px^2
total_skeleton_length_px	34325	px
canalicular_length_density_per_px	0.0340728	px^-1
junction_count	395	count
junction_density_per_px2	0.0003921	px^-2
lacuna_count	12	count
lacunae_per_px2	1.191e-05	px^-2

skeleton_component_count	663	count
mean_component_length_px	51.7722	px
median_component_length_px	21	px
field_length_density_w_per_px	0.0383891	px ⁻¹
field_density_without_flagged_per_px	0.0339923	px ⁻¹
field_density_in_roi_per_px		px ⁻¹

Canalicular width, the new measure

measure	value	units
width_mean_px	6.1251	px
width_median_px	6	px
width_p10_px	4	px
width_p90_px	8	px
width_pixels_used	32892	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel. Measured only on skeleton pixels with real signal under them: bridged pixels, the lacuna buffer and the dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	602	56.2113	21.2037	2.651	0.9261	0.6645	-0.3574	16.2608	914.042	True
2	5924	181.043	48.2611	3.7513	0.9638	0.675	-0.0924	129.871	638.627	False
3	4438	160.681	35.4482	4.5328	0.9754	0.8989	-0.2181	173.351	382.514	False
4	4342	127.796	43.8553	2.914	0.9393	0.893	-0.1983	210.3	874.108	False
5	5254	148.687	46.2758	3.2131	0.9503	0.9227	-0.0688	369.252	79.7008	False
6	3405	124.602	35.4524	3.5146	0.9587	0.8958	-0.0028	532.24	401.315	False
7	4133	149.961	37.557	3.9929	0.9681	0.861	-0.1428	589.071	924.049	False
8	2572	119.294	30.2433	3.9445	0.9673	0.8307	-0.0493	570.032	785.133	False
9	3684	159.337	32.9692	4.8329	0.9784	0.8054	-0.098	739.452	247.536	False
10	3421	128.431	34.9252	3.6773	0.9623	0.8774	-0.2145	813.822	732.524	False
11	2890	110.803	33.408	3.3167	0.9535	0.9299	0.0034	944.816	407.904	False
12	508	44.3947	15.6697	2.8331	0.9356	0.917	-0.2083	990.155	937.699	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	4	196	529	586.688	26	22.5649	5.2001
2	19	574	1322	1962.11	73	26.8783	6.4381
3	8	385	1044	1073.36	33	32.5262	5.8326
4	7	431	1139	961.477	23	41.8034	5.2892
5	6	329	924	290.35	12	24.1959	6.086
6	10	325	1007	1513.65	48	31.5344	5.5614
7	8	473	1313	928.95	37	25.1068	5.7038
8	7	302	892	1068.15	33	32.3682	5.311
9	9	376	978	457.304	17	26.9002	5.2751
10	8	405	1142	1149.72	35	32.8491	5.6559
11	6	230	605	570.487	16	35.6555	4.8394
12	3	98	290	723.198	25	28.9279	4.5752

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	142.811	4575	11641	0.673	0.6207	0.0428415	0.0454428	2.8009
2	625.725	16860	35430	0.9542	0.8861	0.0340451	0.037313	3.0365
3	377.019	13152	31926	1	1	0.0292731	0.0327006	2.1219
4	337.706	12118	29514	1	1	0.0355669	0.0385918	2.0728
5	371.806	13199	31872	1	0.9956	0.0249261	0.028991	1.6137
6	308.593	11379	28385	1	1	0.0285614	0.0354765	3.2405
7	382.877	12760	30245	1	1	0.037069	0.0434121	2.0894
8	306.492	10697	26160	1	1	0.0282322	0.0340979	2.2839
9	437.262	14880	35351	1	1	0.0252688	0.0276654	2.0583
10	326.836	11837	29271	1	1	0.0342148	0.0390147	2.4477
11	263.48	9968	22850	0.9798	0.8783	0.0230738	0.026477	2.2772
12	108.77	5218	13093	0.8837	0.75	0.0187811	0.0221492	2.7581

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	188	296	223.581	596.144	4	7	5
2	523	936	647.884	1481.85	21	18	20
3	231	570	421.563	1170.73	9	13	17
4	222	378	481.09	1305.63	9	15	21
5	170	228	367.534	1041.21	5	11	15
6	254	630	361.049	1145.64	10	15	16
7	302	649	537.73	1477.4	10	21	20
8	192	438	319.409	987.453	8	13	13
9	235	393	422.818	1119.5	10	12	13
10	300	682	453.676	1281.09	12	16	16
11	138	209	254.865	671.227	6	9	6
12	72	165	101.284	311.794	3	3	6

542_z18: distributions, pre-validation, px

